

NONOSCILLATORY INTERPOLATION METHODS APPLIED TO VLASOV-BASED MODELS*

J. A. CARRILLO[†] AND F. VECIL[†]

Abstract. We demonstrate the ability of nonoscillatory interpolation strategies for solving efficiently the transport phase in kinetic systems with applications to charged particle transport in plasmas and semiconductors. Pointwise weighted essentially nonoscillatory (PWENO) interpolation is applied to obtain semi-Lagrangian and flux balance methods that together with splitting techniques form the building blocks of our numerical approach. These methods do not present the restrictive CFL condition typical of finite-difference methods with explicit time-solvers, and, moreover, they provide reliable results controlling parasite oscillations from classical polynomial interpolation while giving highly accurate approximations of smooth parts of the solutions. We perform and compare these methods in different benchmark problems for Vlasov or collisional models for charged particle transport.

Key words. kinetic equations, Vlasov equations, semi-Lagrangian method, flux balance method, Strang's splittings, WENO

AMS subject classifications. 35L60, 35L45

DOI. 10.1137/050644549

1. Introduction.

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f + \frac{F}{m} \cdot \nabla_v f = 0$$

is the basic kinetic model for the description of the motion of charged particles in electronic devices and plasmas under the effect of a force field $F(t, x)$. Here, $f(t, x, v)$ represents the particle number density in phase space (x, v) at time $t > 0$. In order to compute the force field $F = -\nabla_x \Phi$ it is coupled with the Poisson equation

$$\epsilon_0 \Delta_x \Phi = e(\rho[f] - C) = e \left(\int f dv - C \right)$$

for the computation of the electric potential $\Phi(t, x)$ due to both the self-consistent electric potential and an external density $C(x)$, like the doping profile in semiconductors or a background ion density in plasmas. Moreover, when collisional effects have to be taken into account, a Boltzmann operator appears as the right-hand side of the Vlasov equation

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f + \frac{F}{m} \cdot \nabla_v f = \mathcal{Q}[f].$$

One of the main problems while simulating Vlasov-like systems is the presence of violent gradients in the distribution function, which, if not properly controlled by the numerical method, may drive to a false physical description of the phenomena due to

*Received by the editors November 7, 2005; accepted for publication (in revised form) December 22, 2006; published electronically May 29, 2007. The authors acknowledge support from the Spanish DGI-MEC (FEDER) project MTM2005-08024.

<http://www.siam.org/journals/sisc/29-3/64454.html>

[†]ICREA - Departament de Matemàtiques, Universitat Autònoma de Barcelona, 08193 Bellaterra, Spain (carrillo@mat.uab.es, fvecil@mat.uab.es).

a massive formation of spurious oscillations. Several approaches to solving Vlasov-based equations have been proposed in the literature; we refer the reader to [13] for a good review and comparison of different Eulerian solvers. These oscillations appear naturally due to the energy conservation of the system and its Hamiltonian character, the phenomena of the Landau damping being an example of this behavior [30, 13, 14]. When collisions are considered, oscillations are damped, but the control of spurious oscillations continues to be important since high gradients on the moments of the solution (density, mean velocity, and temperature) are typical in the simulations; see [7, 5, 6], for instance, in the semiconductor case.

One of the approaches to attacking the numerical simulation of these models is based on a classical time splitting method, whose theoretical bases were given by Strang [29] and whose application to the Vlasov system goes back to Cheng and Knorr [9]. Time splitting algorithms are used to subdivide a problem into simpler tasks—in our case the transport and the collision steps. In semiconductor devices and plasma simulations, the transport (Vlasov) step in (x, v) - or (x, p) -phase space can be reduced by dimensional splitting to performing advection steps in either dimension. We will elaborate on this later, but let us remark that there are situations in which this dimensional splitting cannot be performed; see [18] for fully relativistic models. Semi-Lagrangian methods were first introduced for meteorology problems [2] and for turbulence study [1]. The application to the Vlasov equation was started by Sonnendrücker et al. in [27]. This method is based on following backward the characteristics, and makes use of some interpolation method: Lagrange polynomials, spline [27], Hermite, or essentially nonoscillatory (ENO) [26]; see [13] for an overview. Flux balance methods were proposed in [14], the third order positive flux conservative method (PFC-3) being a particular case. This method preserves mass and positivity of the solution apart from controlling the total variation of the solution, although it is only third order accurate. This method has been modified in [10] in order to conserve the total energy, too. Several improvements concerning unstructured and adaptive grids and different interpolation techniques have recently been developed in [15, 3].

Another successful approach in the case of collisional semiconductor and plasma models is based on the method of lines with high order nonoscillatory finite-difference reconstructions of the derivatives of the distribution function $f(t, x, v)$ in phase space while performing a total-variation-diminishing (TVD) third order explicit Runge–Kutta method for the time discretization. In this case, spurious oscillations are controlled by ENO [26] or weighted ENO (WENO) [19, 25] reconstructions of the fluxes and thus of the derivatives.

We propose in this paper a semi-Lagrangian (SL) method and a flux balance method (FBM) for Vlasov's systems based on a WENO interpolation for the computation of point values (PWENO); see [24]. We will review this numerical technique for point values reconstruction in section 2.

In section 3, the SL method and the FBM are described. Both methods are based on integrating backward in the characteristics, the second one being mass-conservative and the first one being better as for the control of the total variation. The main advantage with respect to previous semi-Lagrangian methods is the oscillation control while keeping a high order approximation of the solution. Compared to WENO finite-difference methods, our approach avoids the restrictive CFL condition on the time stepping [21] for plasmas and [7, 5, 6] for semiconductors. The proposed numerical method allows the time step Δt to be larger and constant, and the method requires far fewer steps (of an equivalent computational cost) than WENO finite differences. Finally, these numerical techniques could be extended for multidomain or multigrid

computations as done in [24].

In section 4, we show the numerical simulations of five experiments: two pure transport problems and three coupled with collisions. We will elaborate on the advantages and drawbacks of this method in both situations. The exact problems are as follows:

1. A one-dimensional (1D) nonlinear Landau damping [14], where we can observe the filamentation of the phase space during the first phase (strong oscillations in the (x, v) -space) and check whether the method is able to not add noise.
2. A 1D symmetric two stream instability, where we want to observe a vortex appearing after initial time, rotate, and form a typical hole structure.
3. A 1D Vlasov–Boltzmann equation with given confining potential and a linear relaxation operator. Its asymptotic behavior was described in [4, 16].
4. A 1D Vlasov–Fokker–Planck equation with given confining potential. Its asymptotic behavior was described in [11, 17] and it is useful for checking the behavior when variable advection terms appear.
5. A Silicon $n^+ - n - n^+$ -structured semiconductor [8, 7], where collisions are modeled through a linear Boltzmann operator: we compare results given by time splitting methods and finite-difference methods with explicit and implicit-explicit time integrators.

The simulation results demonstrate overall the main features of this numerical approach: no CFL condition, good control of numerical oscillations, and high accuracy of the approximations for transport phases.

2. PWENO interpolations. Pointwise weighted essentially nonoscillatory interpolation methods have been developed in order to compute a direct reconstruction which avoids Lagrange spurious oscillations produced where high derivatives appear.

2.1. Description of the method. We have an interval $[x_L, x_R]$ divided into a grid of N points $x_L = x_0 < x_1 < x_2 < \dots < x_{N-1} = x_R$, and the values of a function f in these points $\{f_i = f(x_i)\}_{i=0, \dots, N-1}$. We choose a stencil of $ntot$ consecutive points from this grid

$$\mathcal{S} = \{x_{first}, \dots, x_{last}\} = \{x_{first}, \dots, x_{first+ntot-1}\}$$

and nlp substencils

$$\mathcal{S}_r = \{x_{last-lpo+1-r}, \dots, x_{last-r}\} = \{x_{i-r}, \dots, x_{last-r}\}$$

for $r = 0, \dots, nlp - 1$, where $lpo := ntot - nlp + 1$ and $i := last - lpo + 1$. Refer to Figure 2.1 for a summary of all the parameters. Let $p_r(x)$ be the Lagrange polynomials of order lpo interpolating f in the stencil $\mathcal{S}_r$:

$$p_r(x) = p_r^{lpo}(x) = \sum_{j=0}^{lpo-1} f_{i-r+j} \prod_{l=0, l \neq j}^{lpo-1} \frac{x - x_{i-r+l}}{x_{i-r+j} - x_{i-r+l}} = \sum_{j=0}^{lpo-1} f_{i-r+j} c_{r,j}(x).$$

The reconstructed value at point x will be a convex combination of the reconstructions at point x given by the nlp Lagrange polynomials in the substencils $\mathcal{S}_r$:

$$p^W(x) = \sum_{r=0}^{nlp-1} \omega_r(x) p_r(x).$$

The weights $\omega_r(x)$ shall give more or less relevance to the stencils depending on how smooth the $p_r(x)$ are, in a way such that the reconstruction is nonoscillatory.

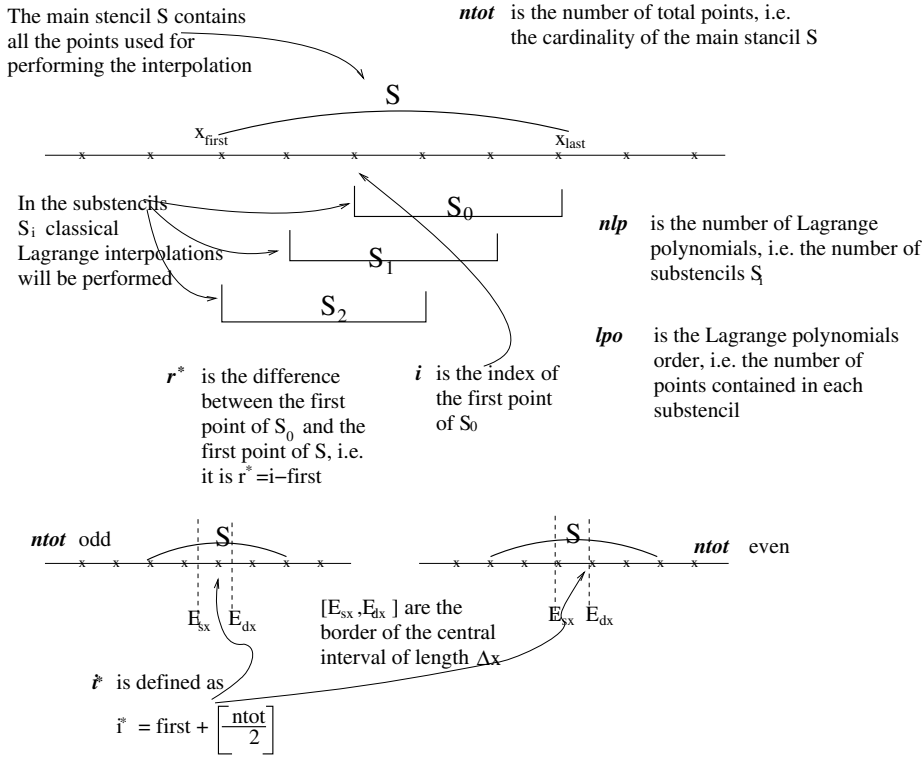


FIG. 2.1. The notation for all the parameters used in the description of the method.

2.2. The smoothness indicators β_r . We shall call smoothness of the polynomial $p_r(x)$ a measure of its derivatives in the interval

$$[\mathcal{E}_L, \mathcal{E}_R] = \begin{cases} [x_{i^* - \frac{1}{2}}, x_{i^* + \frac{1}{2}}] & \text{if } ntot \text{ is odd,} \\ [x_{i^* - 1}, x_{i^*}] & \text{if } ntot \text{ is even,} \end{cases}$$

where the dependencies of $\mathcal{E}_L$ and $\mathcal{E}_R$ will be omitted and we have used the notation

$$i^* = first + \left\lceil \frac{ntot}{2} \right\rceil,$$

in which $[x]$ means the integer part of x . Figure 2.1 graphically shows the meaning of the last notation. If the derivatives are large, the smoothness indicator should be large, and vice versa. The following measurement is proposed by Jiang and Shu in [19]:

$$\beta_r = \sum_{l=1}^{lpo-1} \Delta x^{2l-1} \left\| \frac{d^l}{dx^l} p_r \right\|_{L^2(\mathcal{E}_L, \mathcal{E}_R)}^2.$$

This is a sort of Sobolev norm, where the weights Δx^{2l-1} are needed to make all the terms of the sum independent of Δx , i.e., to make them all be of the same order. Other measurements would be possible, but we shall omit discussing this point.

2.3. The weights $d_r(x)$. In order to achieve a high order reconstruction for regular functions, we define coefficients $d_r(x)$ as

$$(2.1) \quad p[f](x) = \sum_{r=0}^{nlp-1} d_r(x) p_r(x),$$

where $p[f](x)$ is the $ntot$ order Lagrange polynomial interpolating the values of f in the whole stencil $\mathcal{S}$

$$p[f](x) = \sum_{j=0}^{ntot-1} f_{first+j} \prod_{l=0, l \neq j}^{ntot-1} \frac{x - x_{first+l}}{x_{first+j} - x_{first+l}}.$$

Lagrange interpolation gives an $ntot$ order method; by means of these coefficients we want PWENO- $ntot, lpo$ interpolation to approach an $ntot$ order method nonoscillatory for homogeneously regular functions, i.e., whenever all the weights β_r have the same order of magnitude.

PROPOSITION 2.1 (existence and uniqueness of the weights $d_r(x)$). *Let $\mathcal{I} = [x_L, x_R] \subset \mathbb{R}$ be an interval, and let $x_L = x_0 < x_1 < \dots < x_{N-1} = x_R$ be the grid. If x is not a point of the grid, then the weights $d_r(x)$ defined by (2.1) are unique.*

Proof. The $ntot$ order Lagrange interpolation is exact on $\mathbb{P}^{ntot-1}$, i.e.,

$$f \in \mathbb{P}^{ntot-1} \mapsto p[f](x) = f(x),$$

since $p[f]$ is the Lagrange polynomial which interpolates f in $ntot$ points. If $\mathcal{B} = \{b_i\}_{i=0}^{ntot-1}$ is a basis of $\mathbb{P}^{ntot-1}$, $f(x) = \sum_{i=0}^{ntot-1} f^i b_i(x)$ and

$$c_{r,j}(x) = \prod_{l=0, l \neq j}^{lpo-1} \frac{x - x_{i-r+l}}{x_{i-r+j} - x_{i-r+l}};$$

$$(2.2) \quad p[f](x) = p \left[\sum_i f^i b_i \right] (x) = \sum_j \sum_i f^i b_i(x_j) c_{r,j}(x)$$

$$(2.3) \quad = \sum_i f^i \sum_j b_i(x_j) c_{r,j}(x) = \sum_i f^i p[b_i](x),$$

so we need only to impose condition (2.1) on the elements of $\mathcal{B}$. Take as basis

$$\mathcal{B} = \{b_0, b_1, \dots, b_{ntot-1}\} = \left\{ 1, \prod_{l=0}^m (x - x_{first+l}), m \in \{0, \dots, ntot-2\} \right\}.$$

For $deg \leq lpo - 1$, condition (2.1) gives $b_{deg}(x) = \sum_{r=0}^{nlp-1} d_r(x) b_{deg}(x)$, which means

$$\sum_{r=0}^{nlp-1} d_r(x) = 1.$$

For $lpo \leq deg \leq ntot - 1$, condition (2.1) gives

$$b_{deg}(x) = \sum_{r=0}^{nlp-1} d_r(x) p_{r,deg}(x),$$

where $p_{r,deg}(x)$ is the Lagrange polynomial interpolating polynomial $b_{deg}(x)$ at points $\{x_{i-r}, \dots, x_{i-r+lpo-1}\}$, i.e.,

$$p_{r,deg}(x) = \sum_{j=0}^{lpo-1} b_{deg}(x_{i-r+j}) c_{r,j}(x).$$

We get the following linear system:

$$L = \left(\begin{array}{cccc|c} 1 & 1 & \dots & 1 & 1 \\ p_{0,lpo}(x) & p_{1,lpo}(x) & \dots & p_{nlp-1,lpo}(x) & b_{lpo}(x) \\ p_{0,lpo+1}(x) & p_{1,lpo+1}(x) & \dots & p_{nlp-1,lpo+1}(x) & b_{lpo+1}(x) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ p_{0,ntot-1}(x) & p_{1,ntot-1}(x) & \dots & p_{nlp-1,ntot-1}(x) & b_{ntot-1}(x) \end{array} \right).$$

Note that $b_{deg}(x_{i-r+j}) = 0$ for $i-r+j \leq deg-1$, which implies that the matrix has the following appearance:

$$(2.4) \quad L = \left(\begin{array}{cccc|c|c} 1 & 1 & 1 & \dots & 1 & 1 \\ p_{0,lpo}(x) & p_{1,lpo}(x) & \dots & p_{nlp-2,lpo}(x) & 0 & b_{lpo}(x) \\ p_{0,lpo+1}(x) & p_{1,lpo+1}(x) & \dots & 0 & 0 & b_{lpo+1}(x) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ p_{0,ntot-1}(x) & 0 & 0 & \dots & 0 & b_{ntot-1}(x) \end{array} \right).$$

Call $L = (L_1 | L_2)$. The computation of the determinant of L_1 is straightforward:

$$\det(L_1) = - \prod_{s=0}^{nlp-2} p_{s,(ntot-1)-s}(x) \neq 0,$$

which is not zero because none of the terms can be zero. Since

$$p_{s,(ntot-1)-s}(x) = \sum_{j=0}^{lpo-1} b_{(ntot-1)-s}(x_{i-s+j}) c_{s,j}(x)$$

and $b_{(ntot-1)-s}(x_{i-s+j}) = 0$ for $j = 0, \dots, lpo-2$, then

$$p_{s,(ntot-1)-s}(x) = b_{(ntot-1)-s}(x_{i-s+lpo-1}) c_{s,lpo-1}(x) \neq 0$$

because $b_{(ntot-1)-s}(x_{i-s+lpo-1}) \neq 0$; if not, we should get a contradiction: a nonzero polynomial of degree $(ntot-1)-s$ would have $ntot-1$ zeros. $c_{s,lpo-1}(x) \neq 0$ because x does not belong to the points of the grid. In this way, the existence and uniqueness of the $d_r(x)$ have been proven. $\square$

The proof of this proposition, which is quite straightforward, has been reported in full detail since the general code computing any PWENO interpolation, i.e., with any $ntot$ and lpo , uses the linear system (2.4) to compute the weights $d_r(x)$.

2.3.1. Uniqueness of the weights $d_r(x)$ as polynomials. We are able to get an explicit formula for the polynomials $d_r(x)$ by an iterative method. Assume the interpolation points are $\{x_0, x_1, \dots, x_n\}$. Suppose we have two polynomials: $p(x)$ interpolates a function f at the points $x_0, x_1, \dots, x_{n-1}$, and $q(x)$ interpolates at the points $x_1, x_2, \dots, x_n$. A simple exactness argument (Aitken–Neville method) allows

us to check that the polynomial interpolating f at the points $x_0, x_1, \dots, x_n$ is given by

$$r(x) = \frac{p(x)(x - x_n) - q(x)(x - x_0)}{x_0 - x_n}.$$

An explicit recursion procedure based on the previous formula gives us the explicit value of $\tilde{d}_r(x)$. In the case of three points ($n = 2$) we get

$$\begin{cases} \tilde{d}_0(x) = \frac{(x - x_{i-1})(x - x_{i-2})}{(x_{i-1} - x_{i+2})(x_{i-2} - x_{i+2})}, \\ \tilde{d}_1(x) = - \left[\frac{(x - x_{i+2})(x - x_{i-2})}{(x_{i-1} - x_{i+2})(x_{i-2} - x_{i+2})} + \frac{(x - x_{i-2})(x - x_{i+2})}{(x_{i-2} - x_{i+1})(x_{i-2} - x_{i+2})} \right], \\ \tilde{d}_2(x) = \frac{(x - x_{i+1})(x - x_{i+2})}{(x_{i-2} - x_{i+1})(x_{i-2} - x_{i+2})}. \end{cases}$$

PROPOSITION 2.2 (uniqueness of the weights $d_r(x)$). *Let $\mathcal{I} = [x_L, x_R] \subset \mathbb{R}$ be an interval, and let $x_L = x_0 < x_1 < \dots < x_{N-1} = x_R$ be the grid. The weights $d_r(x)$ are unique in $\mathbb{P}^{ntot-lpo}$.*

Proof. Polynomials $\tilde{d}_r(x)$ are explicitly constructed by a standard recursive method based on finite differences, so they exist (no denominator can be zero because $i \neq j \Rightarrow x_i \neq x_j$) and their degree is $(ntot - 1) - (lpo - 1) = ntot - lpo$. Moreover,

$$\forall x \in \mathbb{R} \setminus \{x_i\}_{i=0, \dots, N-1}, \quad \tilde{d}_r(x) = d_r(x)$$

by construction and Proposition 2.1; then the weights $d_r(x)$ are unique as polynomials in $\mathbb{P}^{ntot-lpo}$. $\square$

2.4. The coefficients $\omega_r(x)$. Once we have computed the smoothness indicators, we define

$$(2.5) \quad \tilde{\omega}_r(x) = \frac{d_r(x)}{(\epsilon + \beta_r)^2},$$

where ϵ is just a constant which keeps the denominator from being zero (in the code $\epsilon = 10^{-6}$ is used). To get the weights $\omega_r(x)$ we just have to normalize the $\tilde{\omega}_r(x)$ given by (2.5):

$$\omega_r(x) = \frac{\tilde{\omega}_r(x)}{\sum_{j=0}^{nlp-1} \tilde{\omega}_j(x)}.$$

On one hand, if the function is smooth enough in the stencil $\mathcal{S}$ (all the smoothness indicators are of the same order), then the main role in the coefficients $\omega_r(x)$ is played by the weights $d_r(x)$ making the PWENO reconstruction approach a Lagrange reconstruction, which achieves the best interpolation order for smooth functions. For instance, if all the β_r were equal, then PWENO interpolation would coincide with Lagrange interpolation. On the other hand, if the function has some “irregularity” in the stencil $\mathcal{S}$ (the smoothness indicators are not all of the same order), then the main role in the coefficients $\omega_r(x)$ is played by the smoothness indicators β_r giving more or less weight to the substencils $\mathcal{S}_r$ following the regularity of $p_r(x)$.

2.5. The order of the method. First we recall a standard result about the error committed by Lagrange interpolation (see, for instance, [28]).

PROPOSITION 2.3 (order of Lagrange reconstruction). *Let $\mathcal{I} = [x_L, x_R] \subset \mathbb{R}$ be an interval. If $f \in \mathcal{C}^{n+1}[x_L, x_R]$ and $p \in \mathbb{P}^n$ is the polynomial interpolating f in $n+1$ different points $\{x_0, x_1, \dots, x_n\}$ in $[x_L, x_R]$, then $\forall x \in [x_L, x_R]$ there is a $\xi_x \in]x_L, x_R[$ such that*

$$f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i).$$

In the case of a regular grid this means that

$$p(x) = f(x) + O(\Delta x^{n+1}).$$

LEMMA 2.4. *Assuming that the smoothness indicators satisfy*

$$(2.6) \quad \beta_r = A(1 + O(\Delta x^n)),$$

where A is a nonzero quantity independent of r , then

$$(2.7) \quad \omega_r(x) - d_r(x) = O(\Delta x^n).$$

Proof. Straightforward calculations using Taylor's theorem imply that

$$\frac{1}{(\epsilon + \beta_r)^2} = \frac{1}{(\epsilon + A)^2} + O(\Delta x^n)$$

and

$$\tilde{\omega}_r = \frac{d_r}{(\epsilon + \beta_r)^2} = d_r \left[\frac{1}{(\epsilon + A)^2} + O(\Delta x^n) \right] = \frac{1}{(\epsilon + A)^2} d_r + O(\Delta x^n)$$

with

$$\sum_{j=0}^{nlp-1} \tilde{\omega}_j(x) = \frac{1}{(\epsilon + A)^2} d_r + O(\Delta x^n).$$

Finally, putting together previous estimates, we get

$$\omega_r = \frac{\tilde{\omega}_r}{\sum_{j=0}^{nlp-1} \tilde{\omega}_j(x)} = \frac{\frac{1}{(\epsilon + A)^2} d_r + O(\Delta x^n)}{\frac{1}{(\epsilon + A)^2} + O(\Delta x^n)} = d_r + O(\Delta x^n),$$

giving the desired accuracy result. $\square$

The general error for PWENO interpolation is obtained now.

PROPOSITION 2.5 (order of PWENO reconstruction). *If the weights $d_r(x)$ satisfy*

$$(2.8) \quad \sum_{r=0}^{nlp-1} d_r(x) = 1$$

and $\beta_r = A(1 + O(\Delta x^n))$, where A is a nonzero quantity independent of r , then PWENO- $ntot, lpo$ gives an $(lpo + n)$ order reconstruction.

Proof. First, by Lemma 2.4 we know that (2.7) holds. Since

$$\begin{aligned} p^W(x) - f(x) &= p^W(x) - p(x) + p(x) - f(x) \\ &= \sum_{r=0}^{nlp-1} \omega_r(x) p_r(x) - \sum_{r=0}^{nlp-1} d_r(x) p_r(x) + \sum_{r=0}^{nlp-1} d_r(x) p_r(x) - f(x) \\ &= \sum_{r=0}^{nlp-1} \omega_r(x) p_r(x) - \sum_{r=0}^{nlp-1} d_r(x) p_r(x) + O(\Delta x^{ntot}), \end{aligned}$$

we already know that the method cannot be more than $ntot$ order, and we need to calculate

$$\mathcal{I} = \sum_{r=0}^{nlp-1} \omega_r(x) p_r(x) - \sum_{r=0}^{nlp-1} d_r(x) p_r(x)$$

to check which is the order. By simple manipulations,

$$\begin{aligned} \mathcal{I} &= \sum_{r=0}^{nlp-1} \omega_r(x) p_r(x) - \sum_{r=0}^{nlp-1} d_r(x) p_r(x) = \sum_{r=0}^{nlp-1} [\omega_r(x) - d_r(x)] p_r(x) \\ &= \sum_{r=0}^{nlp-1} [\omega_r(x) - d_r(x)] [p_r(x) - f(x)], \end{aligned}$$

due to (2.8).

$$\mathcal{I} = \sum_{r=0}^{nlp-1} [\omega_r(x) - d_r(x)] [p_r(x) - f(x)] = \sum_{r=0}^{nlp-1} [\omega_r(x) - d_r(x)] O(\Delta x^{lpo})$$

because of Proposition 2.3. We finally deduce

$$\mathcal{I} = \sum_{r=0}^{nlp-1} [\omega_r(x) - d_r(x)] O(\Delta x^{lpo}) = \sum_{r=0}^{nlp-1} O(\Delta x^n) O(\Delta x^{lpo}),$$

due to (2.7). $\square$

Our problem is now translated to the Taylor expansion of the smoothness indicators β_r . Once we have been able to compute n in (2.6), then the method will be order $lpo + n$. By simple manipulations, the smoothness indicators can be expressed as

$$\begin{aligned} \beta_r &= \sum_{l=1}^{lpo-1} \int_{\mathcal{E}} \Delta x^{2l-1} [D^l p_r(x)]^2 dx \\ &= \sum_{j,k} f_{i-r+j} f_{i-r+k} K_{j,k}^r, \end{aligned}$$

where K^r is the symmetric matrix

$$K_{j,k}^r = \sum_{l=1}^{lpo-1} \int_{\mathcal{E}} \Delta x^{2l-1} D^l c_{r,j} D^l c_{r,k} dx.$$

We can now expand $f(x)$ around the point x_i :

$$f(x) = \sum_n \frac{1}{n!} f^{(n)}(x_i) (x - x_i)^n$$

so that

$$f_{i-r+j} = \sum_n \frac{1}{n!} f^{(n)}(x_i) (\Delta x(i - r + j - i))^n = \sum_n \frac{1}{n!} f^{(n)}(x_i) (j - r)^n \Delta x^n.$$

Therefore,

$$\beta_r = \sum_{n,m} \Delta x^{n+m} f^{(n)} f^{(m)} \mathcal{D}_{n,m}^r,$$

where $\mathcal{D}^r$ is the symmetric matrix

$$\mathcal{D}_{n,m}^r = \frac{1}{n!m!} \sum_{j,k=0}^{lpo-1} K_{j,k}^r (j-r)^n (k-r)^m.$$

Computation of these coefficients leads directly to the general order of PWENO methods; for instance, we have the following:

- Numerical computation of $\mathcal{D}^r$ for method WENO-6,4 shows that

$$0 \leq r \leq 2, \quad \mathcal{D}_{1,1}^r = 1, \quad \mathcal{D}_{2,2}^r = \frac{4}{3},$$

and therefore, if $f' \neq 0$, then

$$\beta_r = (f'(x_i) \Delta x)^2 (1 + O(\Delta x))$$

(where $A = (f'(x_i) \Delta x)^2$, $n = 1$ in (2.6)), and if $f'(x_i) = 0$, then

$$\beta_r = \frac{4}{3} (f''(x_i) \Delta x^2)^2 (1 + O(\Delta x))$$

($A = \frac{4}{3} (f''(x_i) \Delta x^2)^2$, $n = 1$ in (2.6)), i.e., that the method is fifth order.

- Computations for WENO-5,3 imply that if $f'(x_i) \neq 0$, then

$$\beta_r = (f'(x_i) \Delta x)^2 (1 + O(\Delta x^2))$$

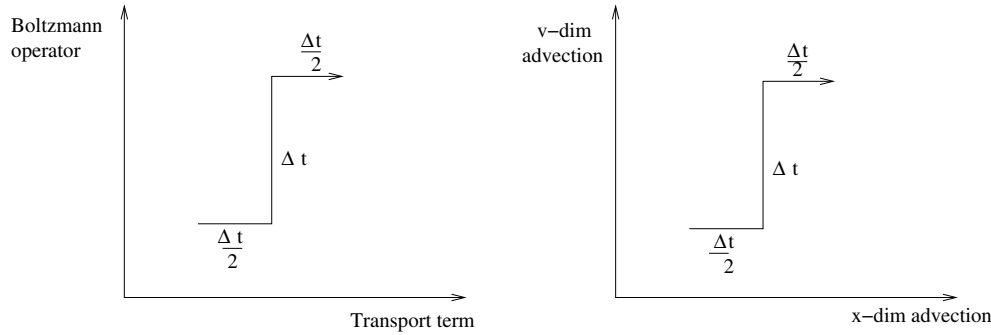
(where $A = (f'(x_i) \Delta x)^2$, $n = 2$ in (2.6)). This implies that WENO-5,3 has order 5.

3. Time splittings and linear advection. For integrating both the Vlasov–Boltzmann coupling

$$\frac{\partial f}{\partial t} + \underbrace{v \cdot \nabla_x f + F \cdot \nabla_v f}_{\text{Vlasov}} = \underbrace{\mathcal{Q}[f]}_{\text{Boltzmann}}$$

and the Vlasov equation

$$\frac{\partial f}{\partial t} + \underbrace{v \cdot \nabla_x f}_{\text{adv. in } x} + \underbrace{F \cdot \nabla_v f}_{\text{adv. in } v} = 0$$

FIG. 3.1. *Time splitting schemes.*

we use Strang [29] and Cheng–Knorr [9] splitting schemes shown in Figure 3.1. In our simulations all reduce to solving the 1D linear advection equation

$$\begin{cases} \frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} = 0, \\ f(t=0, x) = f_0(x), \end{cases}$$

whose solution is the translation with velocity v of the initial function

$$f(t, x) = f_0(x - vt)$$

and is obviously mass-conservative.

Two methods are proposed: the semi-Lagrangian (SL) method and the flux balance method (FBM), the second one being mass-conservative and the first one being better as for controlling oscillations.

3.1. Semi-Lagrangian method. This method is based in directly following backward the characteristics of the system; i.e., knowing

$$f_i^n = f(t^n, x_i),$$

we perform the step in time this way:

$$f_i^{n+1} = f(t^{n+1}, x_i) = f(t^n, x_i - v\Delta t) \simeq p^n(x_i - v\Delta t),$$

where p^n is any chosen interpolation from point values of $f(t^n, \cdot)$. Here, we will choose PWENO as the interpolation method. This method is not mass-conservative.

3.2. Flux balance method. The flux balance method (FBM) is used in [14] to construct a conservative method. We already know that

$$f(t + \Delta t, x) = f(t, x - v\Delta t).$$

Now, let us integrate over an interval $[b_1, b_2]$ to get

$$\begin{aligned} \int_{b_1}^{b_2} f(t + \Delta t, \xi) d\xi &= \int_{b_1}^{b_2} f(t, \xi - v\Delta t) d\xi = \int_{b_1 - v\Delta t}^{b_2 - v\Delta t} f(t, \xi) d\xi \\ &= \int_{b_1 - v\Delta t}^{b_1} f(t, \xi) d\xi + \int_{b_1}^{b_2} f(t, \xi) d\xi - \int_{b_2 - v\Delta t}^{b_2} f(t, \xi) d\xi. \end{aligned}$$

If we use as notation

$$\Psi(t, x) = \int_{x-v\Delta t}^x f(t, \xi) d\xi,$$

we get

$$\int_{b_1}^{b_2} f(t + \Delta t, \xi) d\xi = \int_{b_1}^{b_2} f(t, \xi) d\xi + \Psi(t, b_1) - \Psi(t, b_2),$$

and dividing by $\Delta = b_2 - b_1$,

$$\frac{\int_{b_1}^{b_2} f(t + \Delta t, \xi) d\xi}{\Delta} = \frac{\int_{b_1}^{b_2} f(t, \xi) d\xi}{\Delta} + \frac{\Psi(t, b_1) - \Psi(t, b_2)}{\Delta},$$

which means

$$\bar{f}_{(b_1, b_2)}(t + \Delta t) = \bar{f}_{(b_1, b_2)}(t) + \frac{\Psi(t, b_1) - \Psi(t, b_2)}{\Delta},$$

i.e., the local description of mass-conservation. By denoting by $F(t, \cdot)$ the primitive of $f(t, \cdot)$, the numerical method we get applying the previous argument on the interval $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ is

$$f_i^{n+1} = f_i^n + \frac{\Psi^n(x_{i-\frac{1}{2}}) - \Psi^n(x_{i+\frac{1}{2}})}{\Delta x},$$

where we have approximated the mean value of the function over the interval by its value at the center and

$$\Psi^n(x_{i-\frac{1}{2}}) = \int_{x_{i-\frac{1}{2}} - v\Delta t}^{x_{i-\frac{1}{2}}} f(t^n, \xi) d\xi = F(x_{i-\frac{1}{2}}) - F(x_{i-\frac{1}{2}} - v\Delta t).$$

Finally, $F(x_{i-\frac{1}{2}} - v\Delta t)$ will be approximated by a chosen interpolation from the known point values of F at $x_{i+\frac{1}{2}}$. Here, we will choose PWENO as the interpolation method. A method belonging to this family is the positive flux conservative method (PFC-3) introduced in [14].

3.3. Total variation control. The discrete total variation (DTV) is defined as

$$\sum_{i=0}^{N-2} \|f_i^n - f_{i+1}^n\|.$$

The exact solution of the linear advection equation obviously conserves the total variation. If the numerical method is able to respect this property, then it is not adding spurious oscillations due to the interpolation to the shape of the function. Take as initial datum

$$f^{step}(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x \geq 0 \end{cases}$$

for the linear advection equation with periodic boundary conditions on $x \in [-1, 1]$. As we see in Figure 3.2, Lagrange interpolation has the worst behavior both in the SL method and in the FBM, while PWENO interpolation has a very good conservation of the DTV in the SL method. In the FBM PWENO behaves a little bit worse, but in exchange this second method conserves the mass.

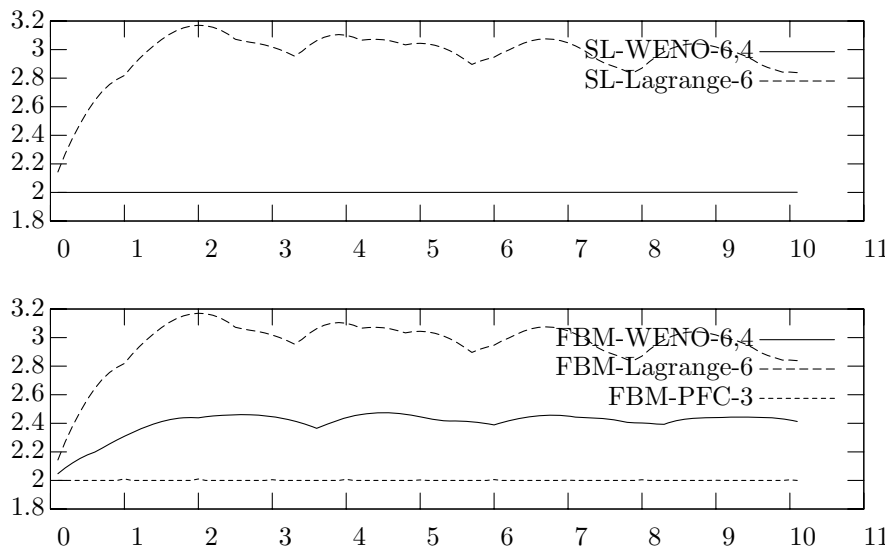


FIG. 3.2. The evolution of DTV against time for the step as initial function.

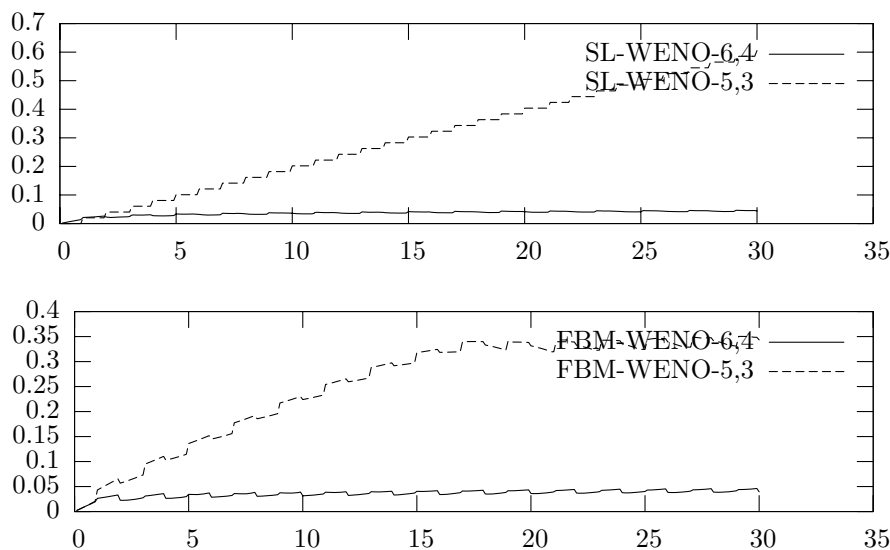


FIG. 3.3. The evolution of L^1 -norm error against the time for $N = 100$, $x \in [-\pi, \pi]$, $\Delta t = 0.1$, $t_{max} = 30$, $f_0(x) = f^{step}(x)$.

3.4. Disphacement errors. Now we want to focus our attention on how important the choices of the substencils, and thus, of the parameters $ntot$ and lpo in the PWENO- $ntot, lpo$ interpolation, are. In Figure 3.3, we see that WENO-6,4 is more accurate than WENO-5,3 both in the semi-Lagrangian method and in the flux balance methods. This difference is due to the fact that in WENO-5,3 not all the substencils “feel” that there is a jump point. Suppose we are interpolating close to x_i , and suppose the jump point is situated between x_i and x_{i+1} . If there is a substencil

which does not contain the irregularity, the WENO method will give weights 0 to all of them but this one. This means that the jump runs with the wrong speed $\frac{\Delta x}{\Delta t}$ with an error after time T with respect to the real speed of $\frac{\Delta x}{\Delta t} [\alpha - 1] \times T$ with $\alpha = v \frac{\Delta t}{\Delta x}$. Then, it is essential to use a WENO method such that every substencil contains any discontinuity which may appear, like WENO-6,4. In general, it must be

$$lpo \geq \left\lceil \frac{ntot + 1}{2} \right\rceil + 1.$$

3.5. Variable advection equation. Previous methods were written for constant advection equations since most of the time they are the ones appearing in Vlasov-like models. However, both the semi-Lagrangian method and the FBM can be applied for transport equations of the type

$$\begin{cases} \frac{\partial f}{\partial t} + \frac{\partial}{\partial x}[a(t, x)f] = 0, \\ f(t = 0, x) = f_0(x), \end{cases}$$

whose solution can be written in terms of the characteristics given by the solution $r(s; t, x)$ of $r'(s) = a(s, r(s))$ with $r(t) = x$ as

$$f(t, x) = f_0(r(0; t, x)) \frac{\partial r}{\partial x}(0; t, x);$$

see also [13]. The semi-Lagrangian approximation is obtained as

$$f_i^{n+1} = f(t^{n+1}, x_i) = f(t^n, r(t^n; t^{n+1}, x_i)) \frac{\partial r}{\partial x}(t^n; t^{n+1}, x_i),$$

where $f(t^n, \cdot)$ is approximated by interpolation if the Jacobian of the flow map is explicitly known. The flux balance method can also be applied in this case without any change since the solution to our transport equation satisfies

$$\int_{b_1}^{b_2} f(t + \Delta t, \xi) d\xi = \int_{r(t; t + \Delta t, b_1)}^{r(t; t + \Delta t, b_2)} f(t, \xi) d\xi$$

for any interval $[b_1, b_2]$ from which the same procedure as in subsection 3.2 applies to obtain the FBM; see also [13].

3.6. Splitting and collisions. Semi-Lagrangian and flux balance methods do not have any CFL limitation as already stressed in the introduction. This is a clear advantage especially when dealing with kinetic equations that possess very large velocities. However, there is an important drawback of using SL methods or FBMs. Due to their “characteristic nature” it is hard to couple them efficiently with a collisional/relaxation right-hand side when this relaxation part is stiff with respect to transport terms. This situation happens in diffusive scalings of kinetic equations, for instance [20]. This is due to the fact that the choice of time integrators is limited to splitting strategies which have a limited accuracy of second order in time away from stiff regions of the right-hand side. The reader should be aware of this potential loss of accuracy induced by the coupling with collisions through splitting.

Finally, let us remark that dimensional splitting for the computation of the transport phases for collisionless Vlasov equations is not always a good choice. Actually, for the Vlasov–Maxwell system for laser-plasma interaction, the instability of a particular

dimensional splitting scheme has been reported in [18]. This system has the feature of having variable advection in both position and momentum and it has to be written in divergence form in phase space leading to a two-dimensional (2D) numerical scheme; we refer the reader to [18] for full details.

4. Numerical simulations. We expose and compare results concerning the simulation of five models:

- A noncollisional coupled Vlasov–Poisson system, by which we study the non-linear Landau damping.
- A noncollisional coupled Vlasov–Poisson system, by which we study the symmetric two stream instability problem.
- A 1D Vlasov system with a linear Boltzmann operator and a given confining potential.
- A 1D Vlasov system with a linear Fokker–Planck operator and a given confining potential.
- A collisional Vlasov–Poisson system, by which a silicon $n^+ - n - n^+$ -structured semiconductor is simulated.

4.1. 1D nonlinear Landau damping. The (normalized) model is

$$\begin{cases} \frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - \frac{\partial \Phi}{\partial x} \frac{\partial f}{\partial v} = 0, \\ \frac{\partial^2 \Phi}{\partial x^2} = 1 - \int_{\mathbb{R}} f dv, \\ f_0(x, v) = f_0(x, v) = \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} \left[1 + 0.5 \cos\left(\frac{x}{2}\right) \right]. \end{cases}$$

Landau theory has been developed in order to study the propagation of small amplitude waves in a uniform plasma (with no magnetic field and no collisions). It conjectures an interchange of energy between the electric field (potential energy) and resonant particles (kinetic energy) driven by the wave, which produces an oscillating evolution of the electric energy

$$\int_0^L |E(t, x)|^2 dx.$$

Its decay is expected to be exponential in the first phase of the evolution, then it should start oscillating around some equilibrium value.

The numerical method must properly take into account two aspects: the filamentation of the phase space and the conservation properties.

The best results are given by FBM-PWENO-5,3: this method is able to describe the filamentation of the phase space; it controls the strong oscillations (part of the physical phenomenon), which, after some time, should start to disappear. It has a good conservation of the total energy (oscillations about 0.03%, 0.015% for Lagrange and 0.05% for PFC-3, as we see in Figure 4.1).

As for FBM-Lagrange-5 and FBM-PFC-3, both give physically unreliable results, the first one because of the spurious growth of the oscillations, and the second one because of the almost total repression of them, as we can see in Figure 4.2. From Figure 4.3 we can see that the reconstruction given by PFC-3 is, in effect, less definite than the other ones, due to the low order. Lagrange reconstruction is more irregular, due to the parasite oscillations.

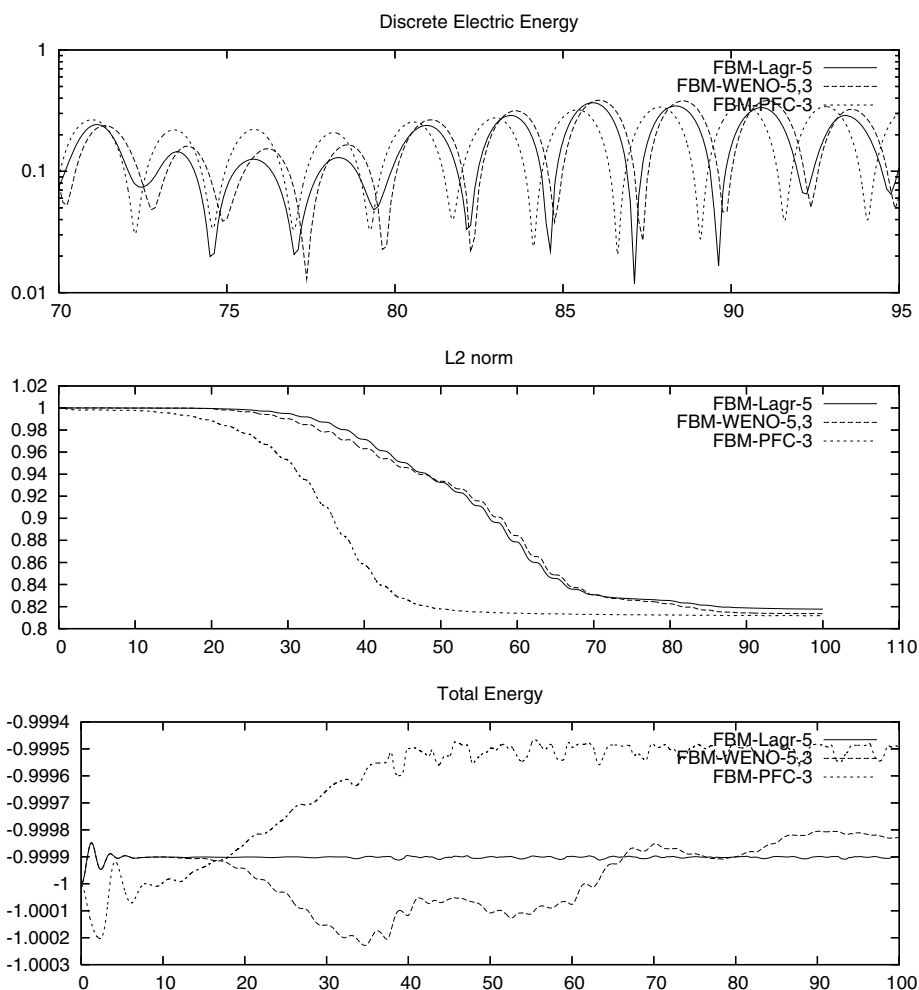


FIG. 4.1. 1D nonlinear Landau damping. Evolution of the discrete electric energy, the L^2 -norm, and the total energy. Test performed with 256×256 points, $\Delta t = 0.125$ for WENO, $\Delta t = 0.01$ for PFC-3, $x \in [0, 4\pi]$, $v \in [-6, 6]$.

4.2. Two stream instability. The (normalized) model is

$$\begin{cases} \frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - \frac{\partial \Phi}{\partial x} \frac{\partial f}{\partial v} = 0, \\ \frac{\partial^2 \Phi}{\partial x^2} = 1 - \int_{\mathbb{R}} f dv, \\ f_0(x, v) = Z \frac{2}{7\sqrt{2\pi}} (1 + 5v^2) \left[1 + \alpha \left(\frac{\cos(2kx) + \cos(3kx)}{1.2} + \cos(kx) \right) \right] e^{-\frac{v^2}{2}} \end{cases}$$

with $\alpha = 0.01$, $k = 0.5$, and Z a normalizing factor to impose periodicity boundary conditions on the electric field. This test has been studied in [13]; there, Filbet and Sonnendrücker compare several methods for a coarse points grid and observe which ones have good conservation properties and which ones are able to follow thin details

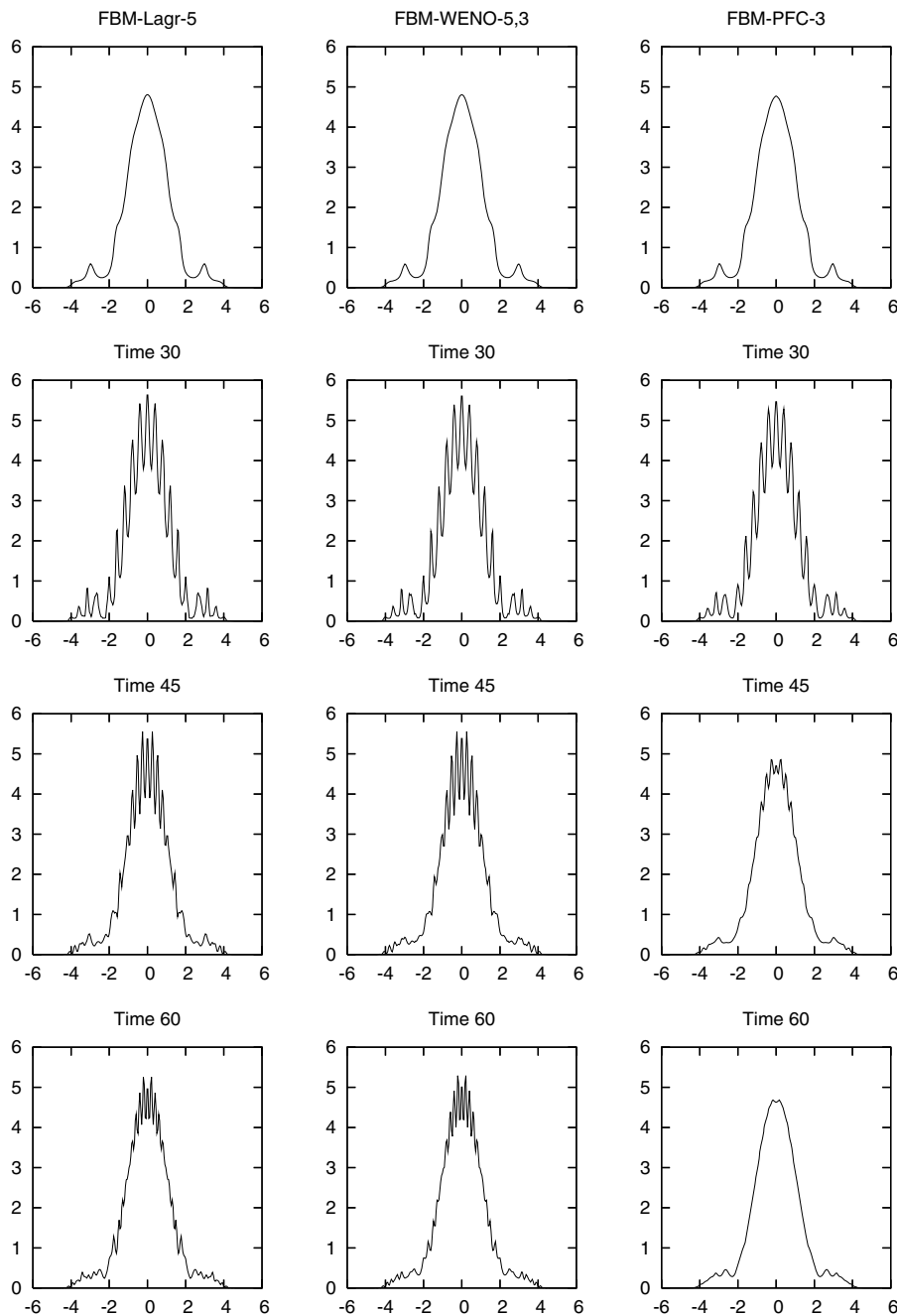


FIG. 4.2. 1D nonlinear Landau damping. Evolution of the x -integrated distribution function for several methods. Test performed with 256×256 points, $\Delta t = 0.125$ for WENO, $\Delta t = 0.01$ for PFC-3, $x \in [0, 4\pi]$, $v \in [-6, 6]$.

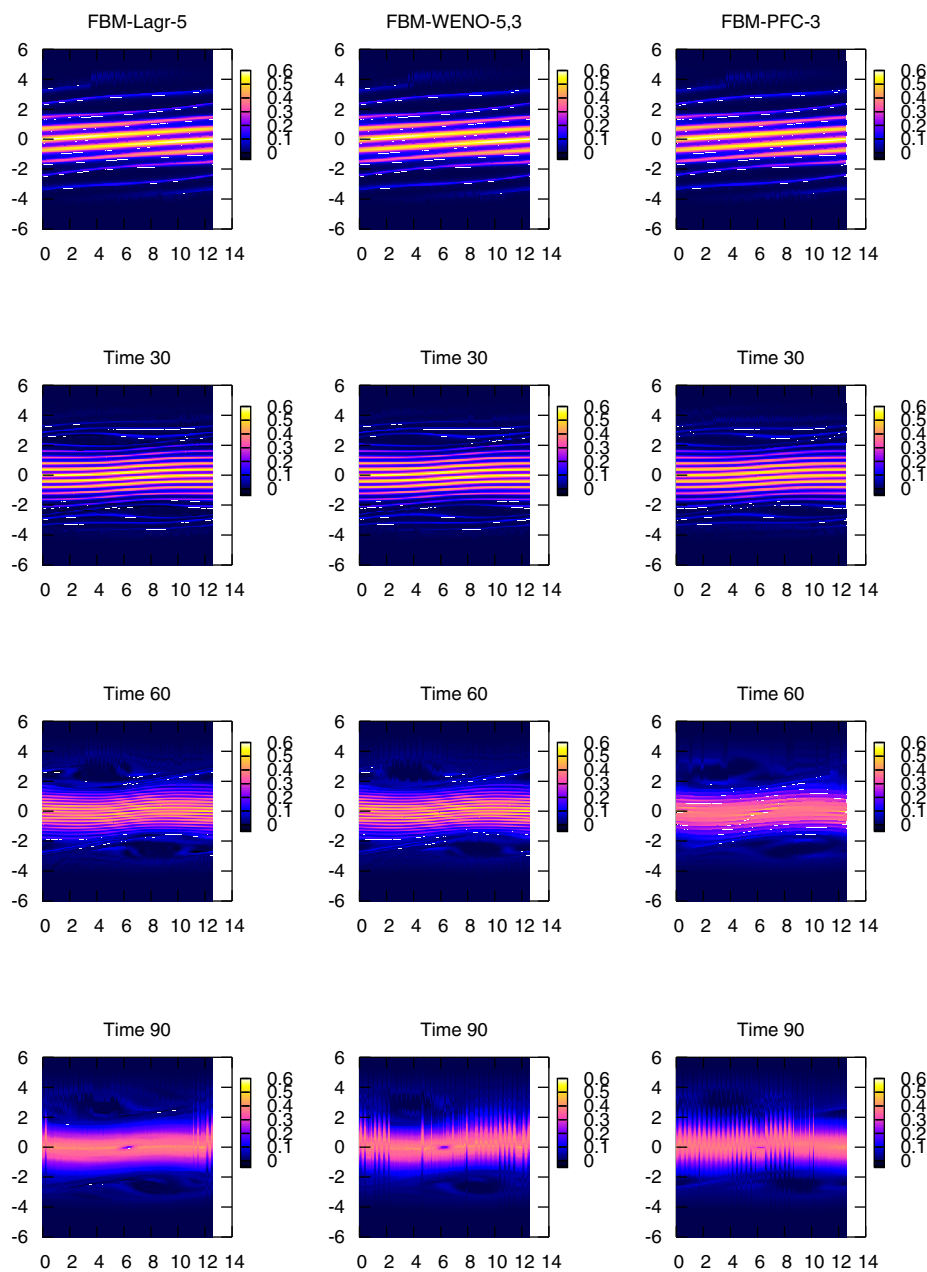


FIG. 4.3. 1D nonlinear Landau damping. Evolution of the level curves of the distribution function for several methods. Test performed with 256×256 points, $\Delta t = 0.125$ for WENO, $\Delta t = 0.01$ for PFC-3, $x \in [0, 4\pi]$, $v \in [-6, 6]$.

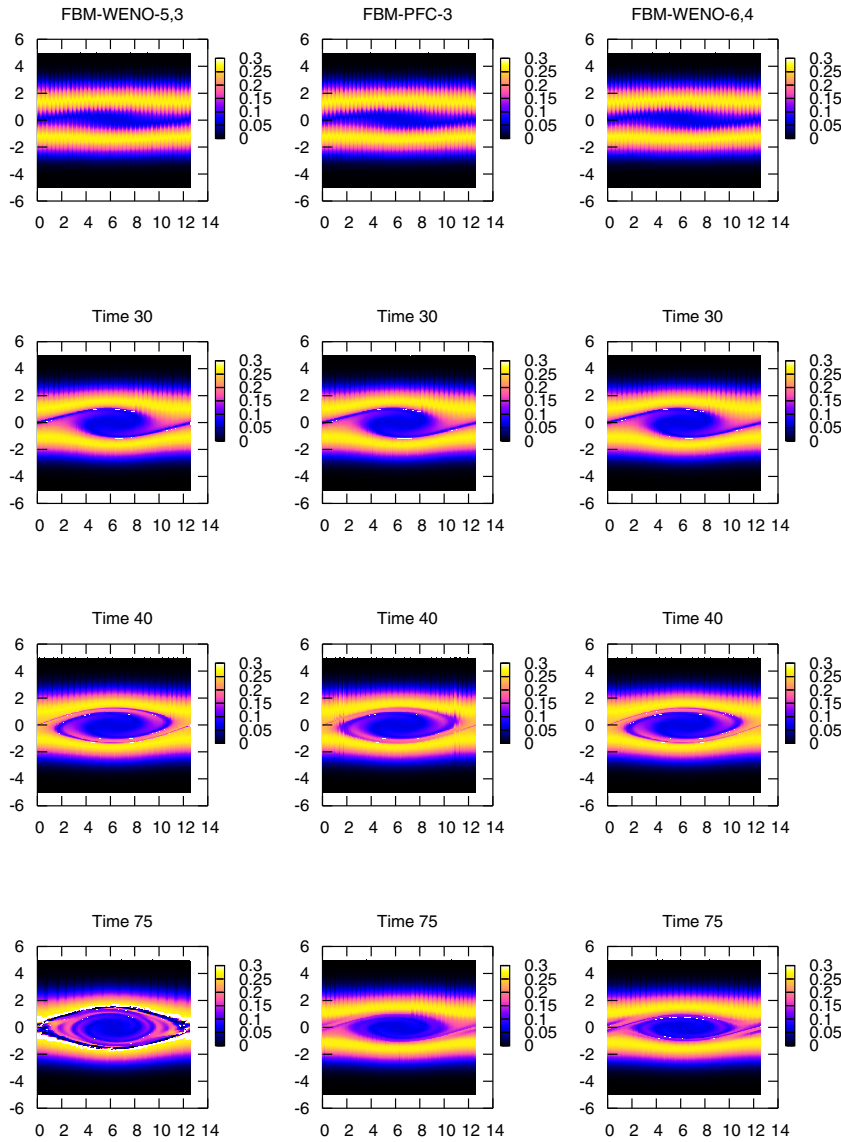


FIG. 4.4. *Two stream instability. Evolution of the level curves. Simulation performed with 256×256 points, $\Delta t = 0.125$ for WENO methods, $\Delta t = 0.01$ for PFC-3, $x \in [0, 4\pi]$, $v \in [-5, 5]$.*

of the distribution function: among their cases, a semi-Lagrangian method with a cubic spline interpolation has given the most reliable results.

Our simulations show that even if both FBM-WENO-5,3 and FBM-WENO-6,4 are fifth order, there may be a substantial difference in their reconstructions, due to the problem the first method has in following shock-like situations, which is the case of this test. Dispersion errors become too important, and in the phase space violent oscillations are produced (look at the level curves in Figure 4.4) which desta-

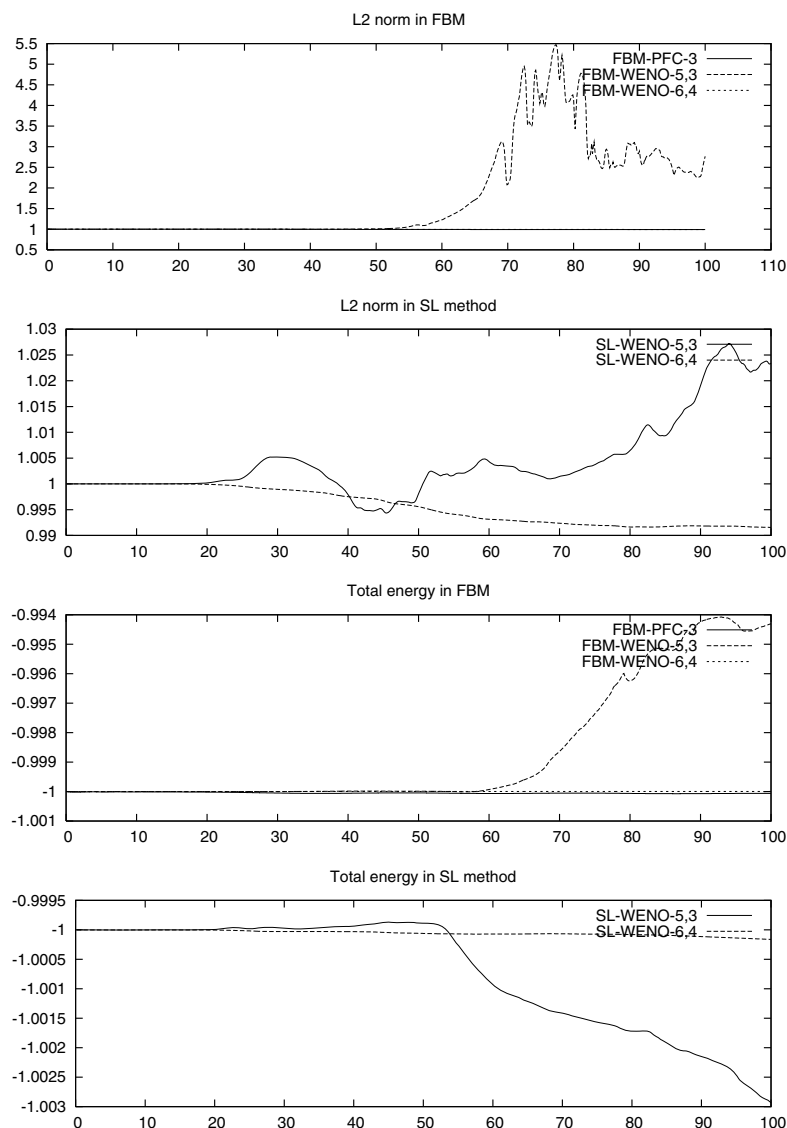


FIG. 4.5. *Two stream instability. Evolution of the L^2 -norm and the total energy. Simulation performed with 256×256 points, $\Delta t = 0.125$ for WENO methods, $\Delta t = 0.01$ for PFC-3, $x \in [0, 4\pi]$, $v \in [-5, 5]$.*

bilize the simulation (the variation of the L^2 -norm is about 400%, as we can remark from Figure 4.5). As for the PFC-3 method, it has good conservation properties (Figure 4.5), but, compared with FBM-WENO-6,4, observing the level curves in Figure 4.4, it is evident that it is lower order and the reconstruction is less detailed.

Let us finally remark that the semi-Lagrangian schemes behave similarly: SL-WENO-5,3 is unstable, too, even if it behaves better than FBM-WENO-5,3; SL-WENO-6,4 as well as FBM-WENO-6,4 gives a fairly detailed reconstruction almost indistinguishable; and the L^2 -norm and the total energy are worse-conserved in the SL method.

4.3. Vlasov–Boltzmann with confining potential. We solve the 1D Vlasov–Boltzmann equation with a confining potential $\Phi_0(x)$ and the simplest linear collision operator:

$$\begin{cases} \frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - \frac{\partial \Phi_0}{\partial x} \frac{\partial f}{\partial v} = \frac{1}{\tau} [\rho M_1 - f], \\ f(0, x) = f_0(x). \end{cases}$$

From [4] we know that the solution tends to a global equilibrium given by

$$f_s = M \left(\int_{\mathbb{R}} \exp(-\Phi_0(x)) dx \right)^{-1} \exp(-\Phi_0(x)) M_1(v)$$

in the L^1 -norm, whenever the external potential verifies the confinement conditions

$$\left\{ \begin{array}{l} \bullet \Phi_0 \geq 0, \Phi_0 \in C^\infty(\mathbb{R}), \\ \bullet \exp(-\Phi_0(x)) \in L^1(\mathbb{R}), \\ \bullet \Phi_0 \text{ is a bounded perturbation of a uniformly convex potential on } \mathbb{R}: \\ \quad \Phi_0 = \Phi_0^{uc} + \Phi_0^{bp} \text{ such that} \\ \quad \text{there exists } \lambda_1 > 0 \text{ such that } \frac{\partial^2}{\partial^2 x} \Phi_0^{uc}(x) \geq \lambda_1 \quad \forall x \in \mathbb{R}, \\ \text{and} \\ \quad \text{there exist } a \text{ and } b \text{ such that } 0 < a \leq \Phi_0^{bp}(x) \leq b < \infty \quad \forall x \in \mathbb{R}. \end{array} \right.$$

The decay rate was proved to be “almost exponential” [4], i.e.,

$$(4.1) \quad \|f - f_s\|_{L^1}^2 \leq H[f; f_s] \leq C(\epsilon, f_0) t^{-\frac{1}{\epsilon}}$$

$\forall \epsilon > 0$. Global and local relative entropies are measures of how far f is from the global equilibrium f_s and the local equilibrium $\rho(t, x)M_1$:

$$(4.2) \quad \begin{cases} H[f; f_s] = \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{|f - f_s|^2}{f_s} dv dx, \\ \tilde{H}[f; \rho M_1] = \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{|f - \rho M_1|^2}{f_s} dv dx. \end{cases}$$

Global and local relative entropies satisfy the ODE inequalities system [4]

$$(4.3) \quad \begin{cases} -\frac{d}{dt} H[f; f_s] \geq K \tilde{H}[f; \rho M_{\theta_0}], \\ \frac{d^2}{dt^2} \tilde{H}[f; \rho M_{\theta_0}] \geq K' H[f; f_s] - C(f, \epsilon) \tilde{H}[f; \rho M_{\theta_0}]. \end{cases}$$

We show numerical results in the particular case of $\Phi_0(x) = \frac{x^2}{2}$. In this case, the Vlasov part gives a rotation of the initial function $f_0(x, v)$ while the collision part thermalizes the velocity distribution toward $M_1(v)$. Moreover, an analysis of the spectrum of operator

$$f \longmapsto -v \frac{\partial f}{\partial x} + x \frac{\partial f}{\partial v} + \frac{1}{\tau} [\rho M_1 - f]$$

has recently been done in [16] showing the existence of a spectral gap in a suitable L^2 -weighted space and, thus, of the exponential convergence in that space and as a consequence in L^1 toward f_s .

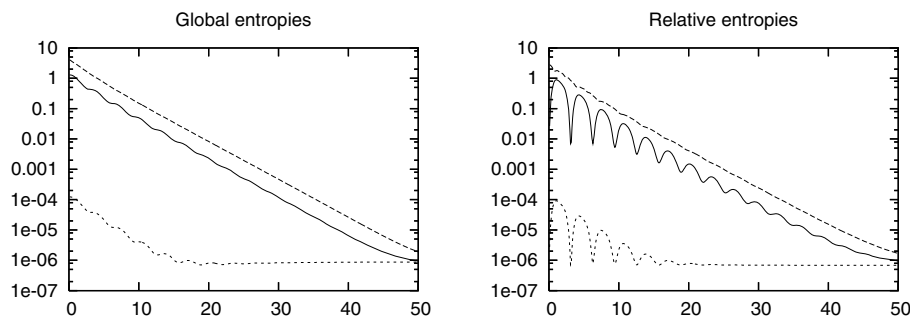


FIG. 4.6. 1D Vlasov–Boltzmann with confining potential. Tests performed with 400×400 points, $\Delta t = 0.1$, SL method with PWENO-6,4, $\tau = 3.5$. Left figure: H -entropy; right figure: $\bar{H}$ -entropy. For the three cases: solid line ($f_0^{(1)}$), dashed line ($f_0^{(2)}$), and small dashed line ($f_0^{(3)}$).

The system of ODE inequalities (4.3) suggests the appearance of oscillations in the trend of solutions toward global equilibrium. The ODE inequalities (4.3) show that the trend of convergence toward local equilibria is compensated by the transport term that should push the solution out of the local equilibria manifold whenever the solution approaches a local equilibrium which is not the global one. This fact may induce an oscillation both in the local and global relative entropies, and thus, on the trend of convergence toward equilibrium.

In Figure 4.6, we compare the decay of global and relative entropies for three different initial functions, two of them far from equilibrium,

$$f_0^{(1)}(x) = Z_1 \sin^2 \left(\frac{x^2}{2} \right) e^{-\frac{x^2+v^2}{2}},$$

$$f_0^{(2)}(x) = Z_2 \sin^2 \left(\frac{x^2}{2} \right) \sin^2 \left(\frac{v^2}{2} \right) e^{-\frac{x^2+v^2}{2}},$$

and a close-to-equilibrium initial data

$$f_0^{(3)}(x) = Z_3 \left[1 + 0.05 \sin^2 \left(\frac{x^2}{2} \right) \right] e^{-\frac{x^2+v^2}{2}}$$

with suitable normalizing factors Z_i , $i = 1, \dots, 3$.

From these log-plot results, we first observe the exponential L^1 -convergence of f toward f_s , as shown in [16], led by the exponential decay of $H[f; f_s]$ due to the first inequality in (4.1). Oscillations of both the global and the local entropies appear. Even if the amplitude of the oscillation is different, stronger for local entropies, it seems evident that the oscillation rate and the decay slope coincide. These oscillations are damped gradually in time; this behavior is observed for the two initial data far from equilibrium, by computing for longer times, while for the close-to-equilibrium initial data this is observed quickly before achieving a steady value due to the accuracy. In [12] oscillations have been reported in the case of the nonlinear Boltzmann equation for rarefied gases in a box with periodic boundary conditions.

4.4. 1D Vlasov–Fokker–Planck with confining potential. We solve the 1D Vlasov equation with a confining potential $\Phi_0(x) = \frac{x^2}{2}$ and a Fokker–Planck operator

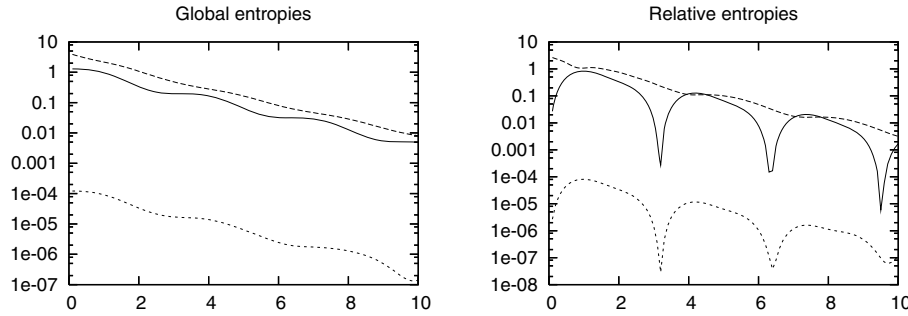


FIG. 4.7. 1D *Vlasov-Fokker-Planck*. Test performed with 400×400 points, $\Delta t = 0.01$, *SL* method with *PWENO*-6,4, $\tau = 3.5$, $\Theta = 1$. Left figure: *H*-entropy; right figure: $\tilde{H}$ -entropy. For the three cases: solid line ($f_0^{(1)}$), dashed line ($f_0^{(2)}$), and small dashed line ($f_0^{(3)}$).

as collision operator:

$$(4.4) \quad \begin{cases} \frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - x \frac{\partial f}{\partial v} = \frac{1}{\tau} \frac{\partial}{\partial v} \left[v f + \Theta \frac{\partial f}{\partial v} \right], \\ f(0, x) = f_0(x). \end{cases}$$

As for the Vlasov-Boltzmann equation, the solution of the above equation tends exponentially fast to a global equilibrium given by

$$f_s = M \left(\int_{\mathbb{R}} \exp \left(-\frac{x^2}{2} \right) dx \right)^{-1} \exp \left(-\frac{x^2}{2} \right) M_{\Theta}(v)$$

in the L^1 -norm. Actually, an almost exponential rate of convergence was obtained in [11] based on a different system of global and local relative logarithmic entropies. In [17] the exponential convergence was shown by spectral theory. In Figure 4.7 we show how far f is from the global equilibrium f_s and the local equilibrium ρM_{Θ} by means of the global and local entropies, respectively, as in the previous test.

Equation (4.4) becomes, after an obvious manipulation,

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - \frac{\partial f}{\partial v} \left[- \left(x + \frac{v}{\tau} \right) f \right] = \frac{\Theta}{\tau} \frac{\partial^2 f}{\partial v^2}.$$

In this case the v -advection velocity is not constant: the term $[-(x + \frac{v}{\tau})f]$ depends on v . Therefore, in order to follow backward the characteristics, we need to solve the ODE $\frac{dv}{dt} = -\frac{v}{\tau} - x$. Applying the procedure of subsection 3.5, the v -advection step with a Δt time step is

$$f(t + \Delta t, x, v) = f \left[t, x, v e^{\Delta t/\tau} - \tau x (e^{\Delta t/\tau} - 1) \right] e^{\Delta t/\tau}.$$

As for the collision step,

$$\frac{\partial f}{\partial t} = \frac{\Theta}{\tau} \frac{\partial^2 f}{\partial v^2}$$

is solved through an explicit Euler step, the second derivative being approximated by centered second order finite difference keeping the standard parabolic CFL condition inside the collision step.

4.5. Simple model of transport in semiconductors. The model is

$$\begin{cases} \frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - \frac{e}{m} E \frac{\partial f}{\partial v} = \frac{1}{\tau} [M_{\Theta_0} \rho[f] - f], \\ \epsilon_{Si} \frac{\partial^2 \Phi}{\partial x^2} = -e [\rho[f] - C], \quad E = -\frac{\partial \Phi}{\partial x}, \\ f_0(x, v) = M_{\Theta_0}(v) C(x), \\ \tau(t, x) = \frac{m}{e} \frac{2\mu_0}{1 + \sqrt{1 + 4((\mu_0/v_0)E(t, x))^2}}, \end{cases}$$

which describes the electron transport in an $n^+ - n - n^+$ diode where the interaction with the semiconductor crystal is taken into account by an effective relaxation time operator. The lattice temperature is set $T_0 = 300K$ and is related to the thermal energy of the lattice by $\Theta_0 = \frac{k_B}{m} T_0$. The relaxation time depends nonlinearly on the electric field producing a saturation on the drift speed; see [7, 8] for more details. We consider a diode of channel length $L = 0.4\mu m$ and doping profile

$$C(x) = \begin{cases} 5 \times 10^5 \frac{1}{\mu m^3}, & 0 \leq x \leq 0.1, \\ 2 \times 10^3 \frac{1}{\mu m^3}, & 0.1 < x < 0.5, \\ 5 \times 10^5 \frac{1}{\mu m^3}, & 0.5 \leq x \leq 0.6. \end{cases}$$

We apply a potential drop at the drain, $\Phi(t, 0) = 0$ V, and $\Phi(t, L) = V_{bias}$ V. Concerning boundary conditions at $x = 0$ and $x = L$, we use inflow boundary conditions fixed by Maxwellians with the lattice temperature for the ghost points outside the physical grid needed for computing PWENO interpolation.

In Figure 4.8, density, mean velocity, electric field, potential, energy, and the distribution function in v at a point near the end of the channel at the stationary state are plotted. We are comparing the results of our time splitting method based on FBM-WENO-6,4 with time step $\Delta t = 0.01$, finite-difference WENO5 coupled with an explicit adaptive in time third order Runge–Kutta (RK3) scheme for time discretization [7, 8] and finite-difference WENO5 coupled with an implicit (collisions)-explicit (transport) Runge–Kutta (IMEX-RK) LRR(3, 2, 2) scheme for time discretization [22, 23] with time step $\Delta t = 0.001$. Results of three simulations are almost indistinguishable; the time step of the explicit RK3 is $\Delta t \simeq 0.0005$ chosen by the CFL due to the explicit discretization of the transport terms.

In Figure 4.9, we compare the results given by the explicit RK3 finite-difference WENO5 method and by FBM-WENO-6,4 for several time steps. One of the main advantages of SL or FBM-WENO schemes is that they do not have an a priori restriction on the time step as the CFL condition for finite-difference WENO5. Therefore, our scheme is much less costly from the computational point of view. Actually, 200 time steps corresponding to $\Delta t = 0.05$ of FBM-WENO-6,4 are enough to give results with less than 4% L^∞ -difference in density to 9000 time steps corresponding to $\Delta t \simeq 0.00075$ of the WENO5 method with explicit RK3 time integrator with comparable computational cost for each time step between the methods. On the other hand, WENO5 with IMEX-RK time integrator with $\Delta t \simeq 0.003$, 2250 time steps, gives comparable results, less than 4% L^∞ -difference in density, while the cost is higher since at each time step one linear system of the order of the velocity grid per position grid point has to be solved.

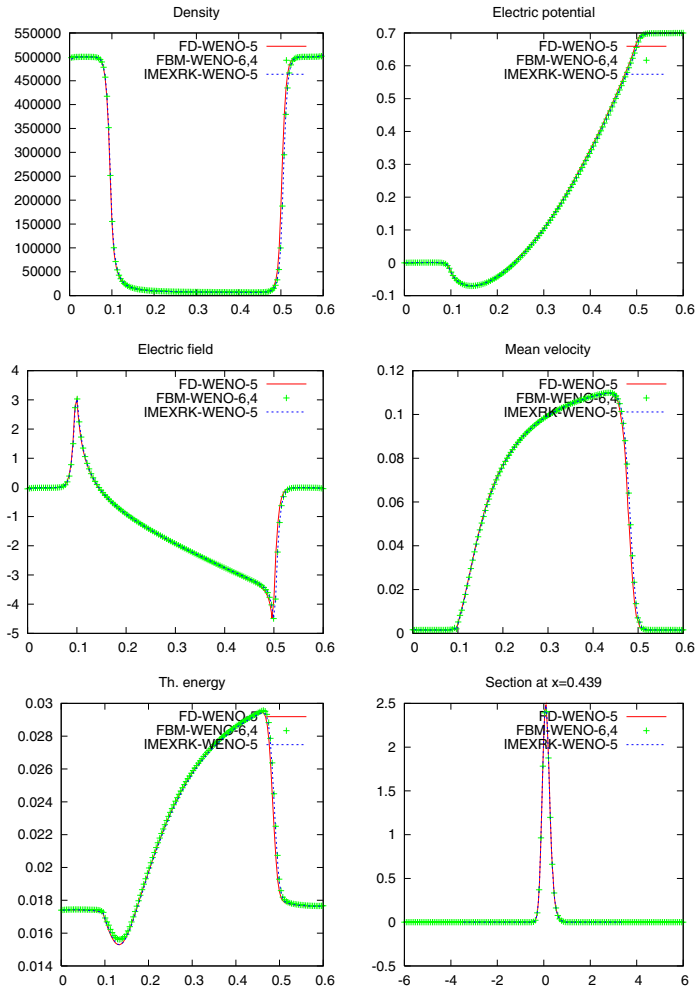


FIG. 4.8. 1D $n^+ - n - n^+$ Silicon semiconductor. Evolution of several macroscopic magnitudes, for explicit Runge-Kutta-3 WENO-5 finite-difference method (CFL is set to 0.3), WENO-6,4 flux balance method with $\Delta t = 0.01$, and an IMEX-LRR(3, 2, 2) WENO-5 finite-difference method with $\Delta t = 0.001$. Grids are set 150×150 points.

Finally, we show in the next table the maximal time step needed to get such 4% L^∞ -relative difference between density for different values of the temperature:

Temperature	Maximal Δt for densities
300	0.05
273	0.05
177	0.04
77	0.01

This table together with the behavior of the L^1 -relative difference for different temperatures observed on the right in Figure 4.9 shows the problem gets stiffer as the temperature goes down and the mesh has to be taken finer. We summarize the phys-

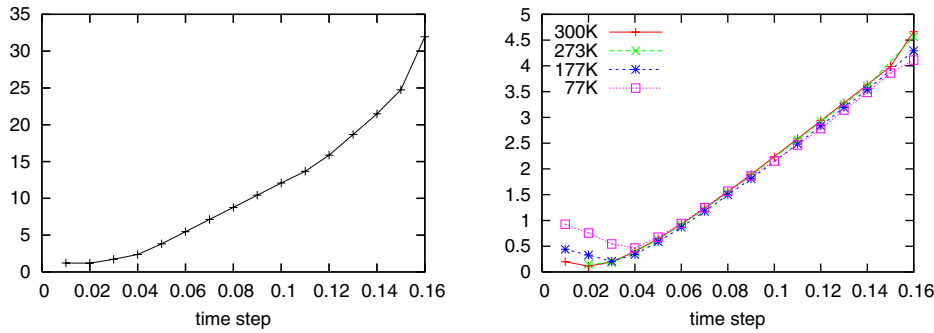


FIG. 4.9. Left: L^∞ -relative difference in % between densities given by the explicit RK3 finite-difference WENO5 method and by the FBM-WENO-6,4 for several time steps. Right: L^1 -relative difference in % between densities given by the explicit RK3 finite-difference WENO5 method and by the FBM-WENO-6,4 for several time steps and temperatures. All runs are with a mesh 150×150 .

ical constants used for the simulation:

Constant	Physical meaning	Magnitude
m	effective electron mass	$0.26 \times 0.9109(10^{-30} Kg)$
e	elementary electric charge	$0.1602(10^{-18} C)$
k_b	Boltzmann's constant	$0.138046 \times 10^{-4}(10^{-18} \frac{J}{K})$
ϵ_{Si}	Silicon dielectric permittivity	$11.7 \times 8.85418(10^{-18} \frac{F}{\mu m})$
μ_0	bulk mobility	0.1323
v_0	saturation velocity	0.13.

5. Conclusions. We have proposed a general family of methods based on essentially nonoscillatory interpolations (PWENO) for solving transport phases of Vlasov-like systems. Dimensional splitting has been used to reduce transport phases to 1D advection. The 1D advection is solved by semi-Lagrangian or flux balance methods coupled with the PWENO interpolation. Interpolation methods that feel all candidate stencils perform better in shock-like situations, and due to mass conservation, we take FBM-WENO-6,4 as the preferred method for transport phases. The performances of these methods have been shown in the detailed list of numerical experiments both for collisionless systems and nonstiff collisional systems where coupling has been done through time-splitting. Since the coupling of collisions and transport is reduced to splitting techniques of low order in time due to the characteristic nature of the transport schemes, the reader should be aware of the potential loss of accuracy when coupling this kind of method with stiff collisional terms.

Acknowledgments. We thank Simon Labrunie for fruitful comments and discussions and José M. Mantas for his help with the IMEX-RK method and general purpose codes applied to the last test.

REFERENCES

- [1] R. ABGRALL AND C. BASDEVANT, *A semi-Lagrangian numerical scheme for two-dimensional turbulence*, C. R. Acad. Sci. Paris Sér. I Math., 305 (1987), pp. 315–318.
- [2] J. R. BATES, *Semi-Lagrangian advective schemes and their use in meteorological modeling*, in Large-Scale Computations in Fluid Mechanics, Part 1 (La Jolla, CA, 1983), Lectures in Appl. Math. 22-1, AMS, Providence, RI, 1985, pp. 1–29.

- [3] N. BESSE AND E. SONNENDRÜCKER, *Semi-Lagrangian schemes for the Vlasov equation on an unstructured mesh of phase space*, J. Comput. Phys., 191 (2003), pp. 341–376.
- [4] J. A. CARRILLO, M. J. CÁCERES, AND T. GOUDON, *Equilibration rate for the linear inhomogeneous relaxation-time Boltzmann equation for charged particles*, Comm. Partial Differential Equations, 28 (2003), pp. 969–989.
- [5] J. A. CARRILLO, I. M. GAMBA, A. MAJORANA, AND C. W. SHU, *A WENO-solver for the transients of Boltzmann–Poisson system for semiconductor devices. Performance and comparisons with Monte Carlo methods*, J. Comput. Phys., 184 (2003), pp. 498–525.
- [6] J. A. CARRILLO, I. M. GAMBA, A. MAJORANA, AND C. W. SHU, *2D semiconductor device simulations by WENO–Boltzmann schemes: Efficiency, boundary conditions, and comparison to Monte Carlo methods*, J. Comput. Phys., 214 (2006), pp. 55–80.
- [7] J. A. CARRILLO, I. GAMBA, AND C. W. SHU, *Computational macroscopic approximations to the one-dimensional relaxation-time kinetic system for semiconductors*, Phys. D, 146 (2000), pp. 289–306.
- [8] C. CERCIGNANI, I. GAMBA, J. JEROME, AND C. W. SHU, *Device benchmark comparisons via kinetic, hydrodynamic, and high-field models*, Comput. Methods Appl. Mech. Engrg., 181 (2000), pp. 381–392.
- [9] C. Z. CHENG AND G. KNORR, *The integration of the Vlasov equation in configuration space*, J. Comput. Phys., 22 (1976), pp. 330–348.
- [10] N. CROUSEILLES AND F. FILBET, *Numerical approximation of collisional plasmas by high order methods*, J. Comput. Phys., 201 (2004), pp. 546–572.
- [11] L. DESVILLETES AND C. VILLANI, *On the trend to global equilibrium in spatially inhomogeneous entropy-dissipating systems: The linear Fokker–Planck equation*, Comm. Pure Appl. Math., 54 (2001), pp. 1–42.
- [12] F. FILBET, C. MOUHOT, AND L. PARESCHI, *Solving the Boltzmann equation in $N \log_2 N$* , SIAM J. Sci. Comput., 28 (2006), pp. 1029–1053.
- [13] F. FILBET AND E. SONNENDRÜCKER, *Comparison of Eulerian Vlasov solvers*, Comput. Phys. Comm., 150 (2003), pp. 247–266.
- [14] F. FILBET, E. SONNENDRÜCKER, AND P. BERTRAND, *Conservative numerical schemes for the Vlasov equation*, J. Comput. Phys., 172 (2001), pp. 166–187.
- [15] M. GUTNIC, M. HAEFELE, I. PAUN, AND E. SONNENDRÜCKER, *Vlasov simulations on an adaptive phase-space grid*, Comput. Phys. Comm., 164 (2004), pp. 214–219.
- [16] F. HÉRAU, *Hypercoercivity and exponential time decay for the linear inhomogeneous relaxation Boltzmann equation*, Asymptot. Anal., 46 (2006), pp. 349–359.
- [17] F. HÉRAU AND F. NIER, *Isotropic hypoellipticity and trend to equilibrium for the Fokker–Planck equation with high degree potential*, Arch. Ration. Mech. Anal., 171 (2004), pp. 151–218.
- [18] F. HUOT, A. GHIZZO, P. BERTRAND, E. SONNENDRÜCKER, AND O. COULAUD, *Instability of the time splitting scheme for the one-dimensional and relativistic Vlasov–Maxwell system*, J. Comput. Phys., 185 (2003), pp. 512–531.
- [19] G. S. JIANG AND C. W. SHU, *Efficient implementation of weighted ENO schemes*, J. Comput. Phys., 126 (1996), pp. 202–228.
- [20] S. JIN, L. PARESCHI, AND G. TOSCANI, *Uniformly accurate diffusive relaxation schemes for multiscale transport equations*, SIAM J. Numer. Anal., 38 (2000), pp. 913–936.
- [21] S. LABRUNIE, J. A. CARRILLO, AND P. BERTRAND, *Numerical study on hydrodynamic and quasi-neutral approximations for collisionless two-species plasmas*, J. Comput. Phys., 200 (2004), pp. 267–298.
- [22] L. PARESCHI AND G. RUSSO, *Implicit-explicit Runge-Kutta schemes for stiff systems of differential equations*, in Recent Trends in Numerical Analysis, L. Brugnano and D. Trigiante, eds., Adv. Theory Comput. Math. 3, Nova Science Publishers, Commack, NY, 2000, pp. 269–288.
- [23] L. PARESCHI AND G. RUSSO, *Implicit-explicit Runge-Kutta methods and applications to hyperbolic systems with relaxation*, J. Sci. Comput., 25 (2005), pp. 129–155.
- [24] K. SEBASTIAN AND C. W. SHU, *Multidomain WENO finite difference method with interpolation at subdomain interfaces*, J. Sci. Comput., 19 (2003), pp. 405–438.
- [25] C.-W. SHU, *Essentially non-oscillatory and weighted essentially non-oscillatory schemes for hyperbolic conservation laws*, in Advanced Numerical Approximation of Nonlinear Hyperbolic Equations, Lecture Notes in Math. 1697, Springer-Verlag, Berlin, 1998, pp. 325–432.
- [26] C. W. SHU AND S. OSHER, *Efficient implementation of essentially non oscillatory shock capturing schemes*, J. Comput. Phys., 77 (1988), pp. 439–471.
- [27] E. SONNENDRÜCKER, J. ROCHE, P. BERTRAND, AND A. GHIZZO, *The semi-Lagrangian method for the numerical resolution of the Vlasov equation*, J. Comput. Phys., 149 (1999), pp. 201–220.

- [28] J. STOER AND R. BULIRSCH, *Introduction to Numerical Analysis*, Springer-Verlag, New York, 1993.
- [29] G. STRANG, *On the construction and comparison of difference schemes*, SIAM J. Numer. Anal., 5 (1968), pp. 506–517.
- [30] T. ZHOU, Y. GUO, AND C. W. SHU, *Numerical study on Landau damping*, Phys. D, 157 (2001), pp. 322–333.